



York Region Crime Analysis: A Data-Driven Plan for 2026

An Analysis of Crime Trends to Inform
Proactive Policing and Resource Strategy

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- Crime Trends & Patterns
- Diagnosing the 2024 Crisis
- Forecasting the Future Threat for 2026
- Spotlight on District 4 (Vaughan)
- Our Strategic Action Plan



Data Foundation

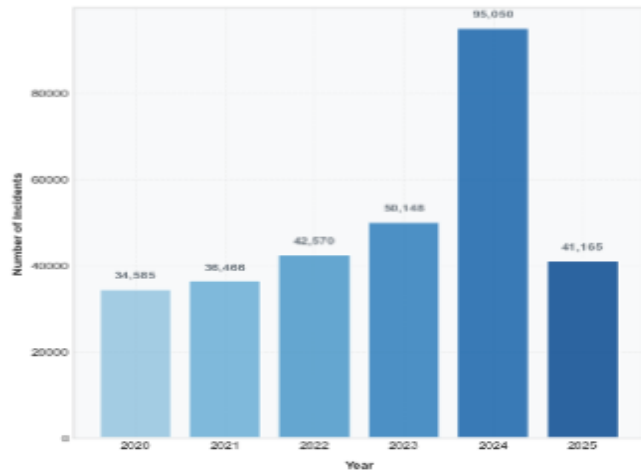
- **Crime Data**
- **911 Calls**
- **Officer On/Off Duty Data**
- **External Data**
 - Focused on factors with a plausible, documented correlation to crime or police workload
 1. **Weather data** (temperature), as higher temperatures can correlate with increased outdoor activity and certain crime types
 2. More critically, **holiday calendars and road safety data** to account for predictable surges in non-crime call volumes (like traffic accidents) that still tie up officer resources
 - Data/feature selection: XGBoost model to calculate feature importance, ensuring the external data added predictive value



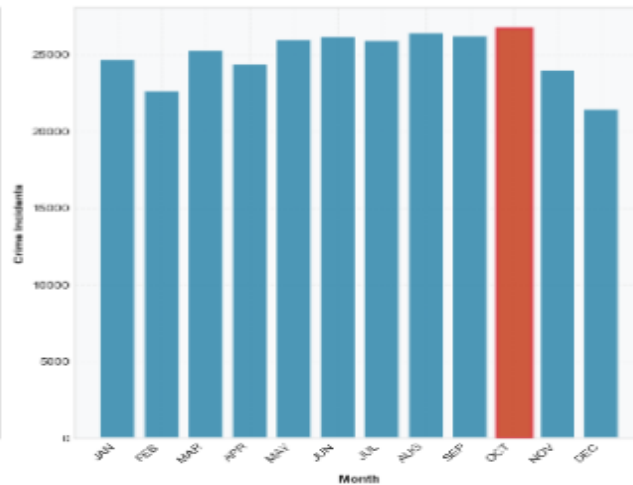
Crime Trend (Temporal Patterns)

YORK REGION CRIME ANALYTICS DASHBOARD
Temporal Patterns and Trends Analysis

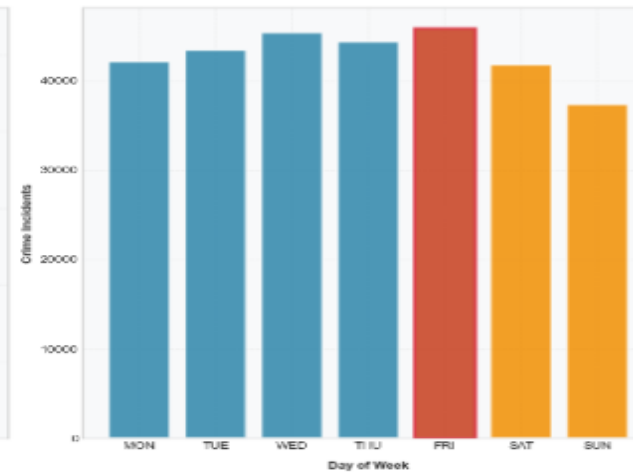
ANNUAL CRIME TRENDS



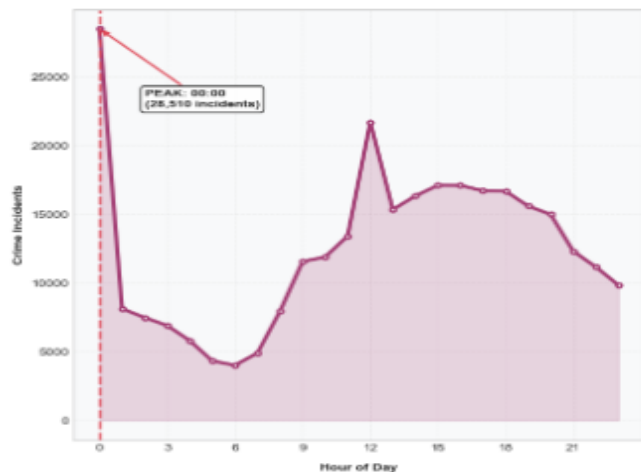
MONTHLY DISTRIBUTION PATTERN



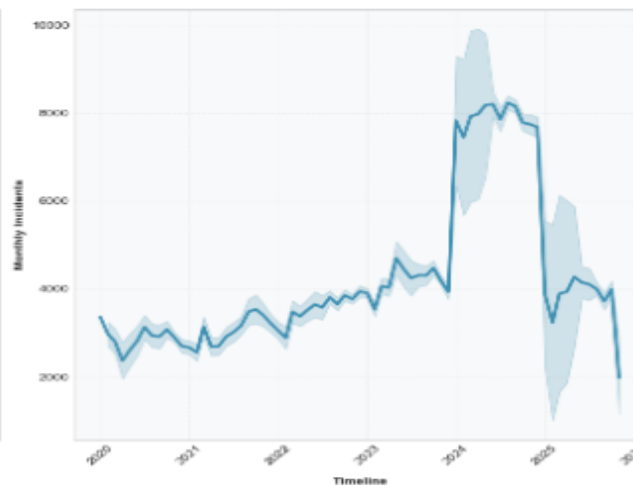
WEEKLY PATTERN ANALYSIS



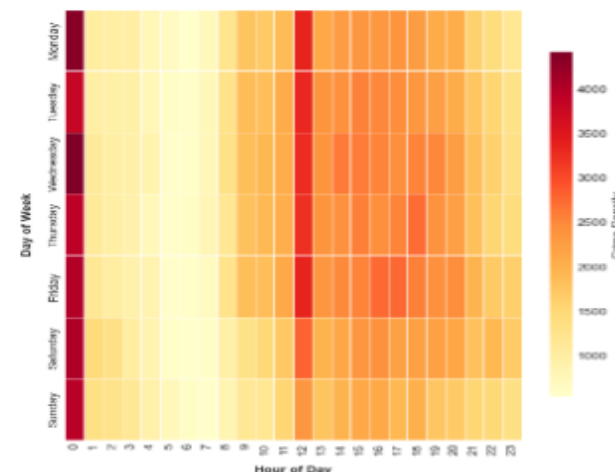
HOURLY DISTRIBUTION



MONTHLY TREND ANALYSIS



CRIME DENSITY HEATMAP
Hour vs Day of Week



Strategic insights

- Weekend Top Activity
- Evening Peak
- High Season:
August, September, October
- Low Season:
February, November, December

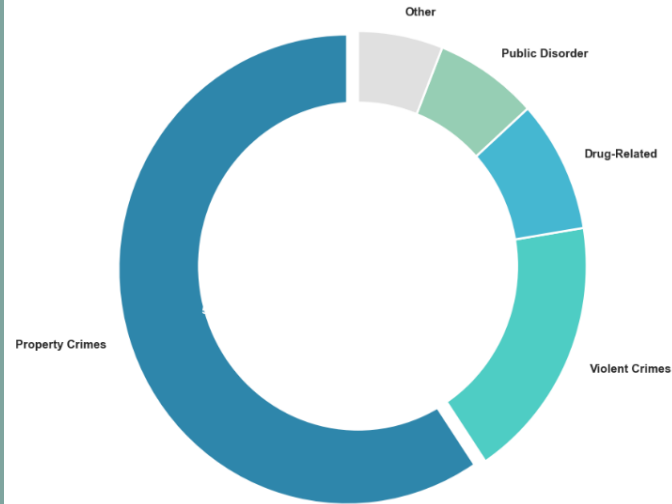
Implication

Crime is predictable. It has a rhythm—peaking on summer evenings and weekends. We can and must use this predictability to our advantage by pre-deploying resources, which is the foundation of our 'Summer Surge Plan'

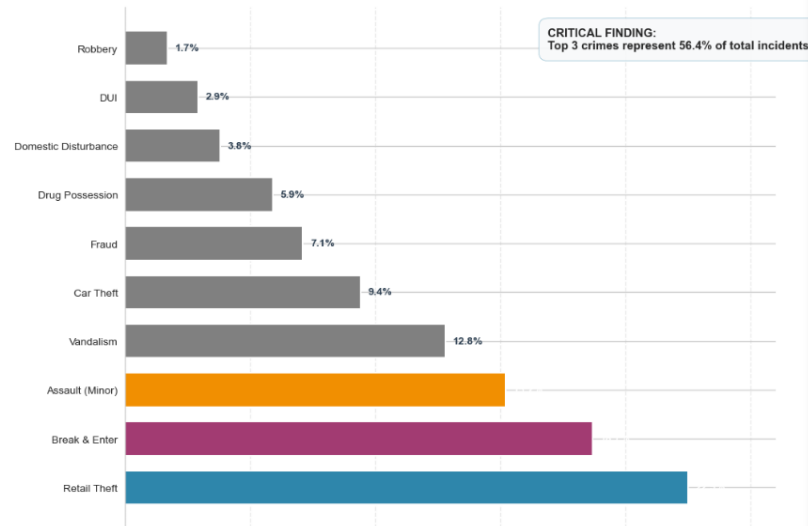


Crime Category Distribution

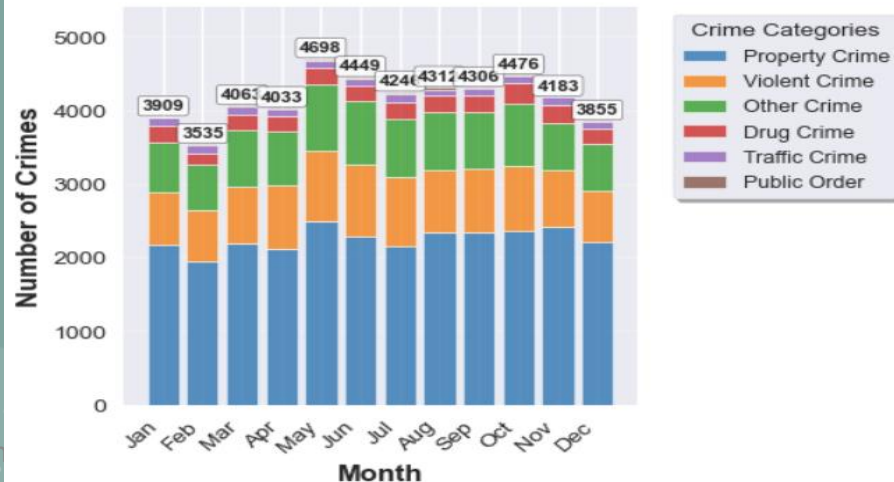
CRIME CATEGORY DISTRIBUTION
Property Crimes Represent 59.4% of Total Incidents



TOP 10 CRIME TYPES
Incidents Driving Police Workload



Monthly Crime Distribution by Category
2023 Analysis



Seasonal Crime Patterns
Top Categories



- Property and Violent Crime Dominate Volume
- Core Message:**
Property and Violent crime are not just the most common; they are growing at an alarming rate. This isn't a general crime problem; it's a specific one requiring category-specific tactics, especially in auto theft and gun violence



The Vicious Cycle of Crime and Staffing

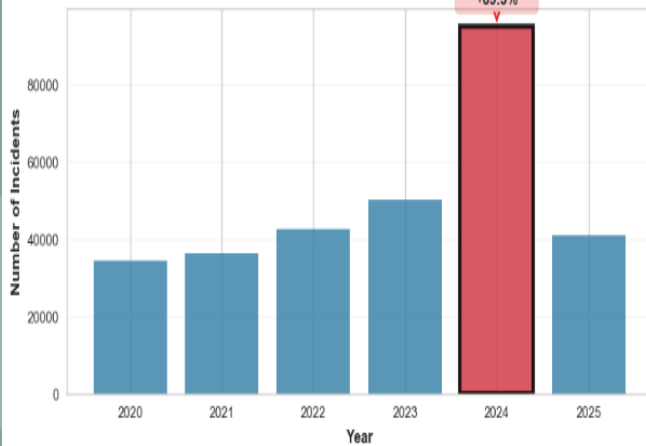
- The Vicious Cycle: High Crime → 911 Calls ↑ → Overtime & Burnout ↑ → Staffing Shortages ↑ → Response Capacity ↓
- High crime volumes create a feedback loop that depletes officer capacity, reducing our ability to respond and prevent future crime, particularly in District 4.



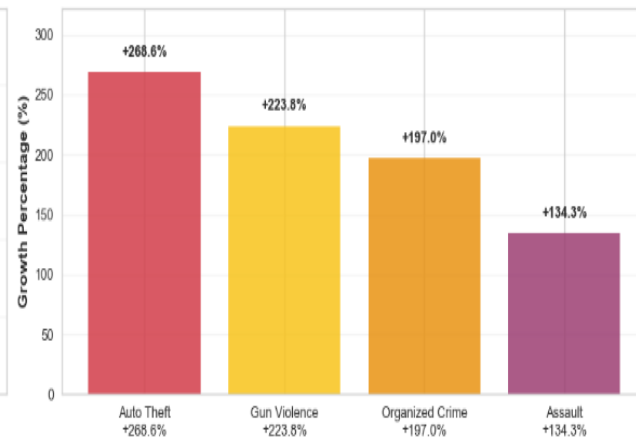


Yearly Crime Trends & The 2024 Anomaly

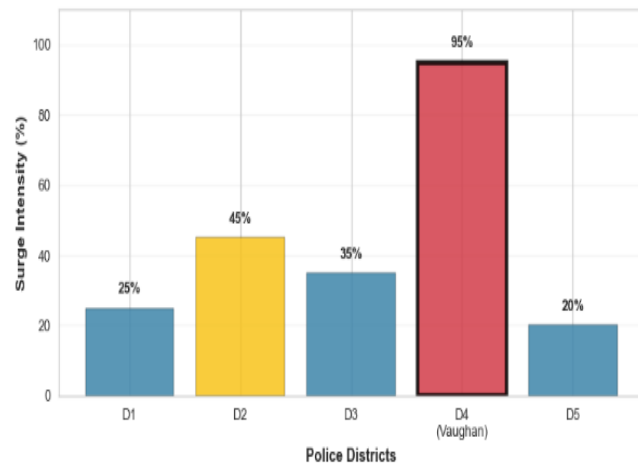
ANNUAL CRIME TRENDS



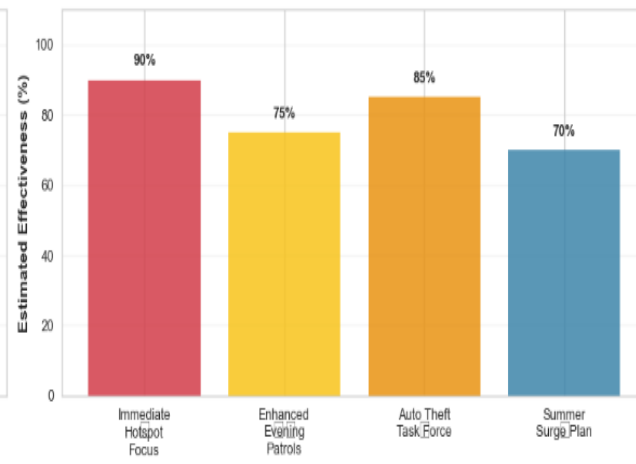
TOP 4 DRIVERS: GROWTH RATES
2024 vs 2023 Comparison



GEOGRAPHIC SURGE INTENSITY BY DISTRICT
District 4: Primary Hotspot



RECOMMENDED RESPONSE STRATEGY
Effectiveness & Urgency Assessment



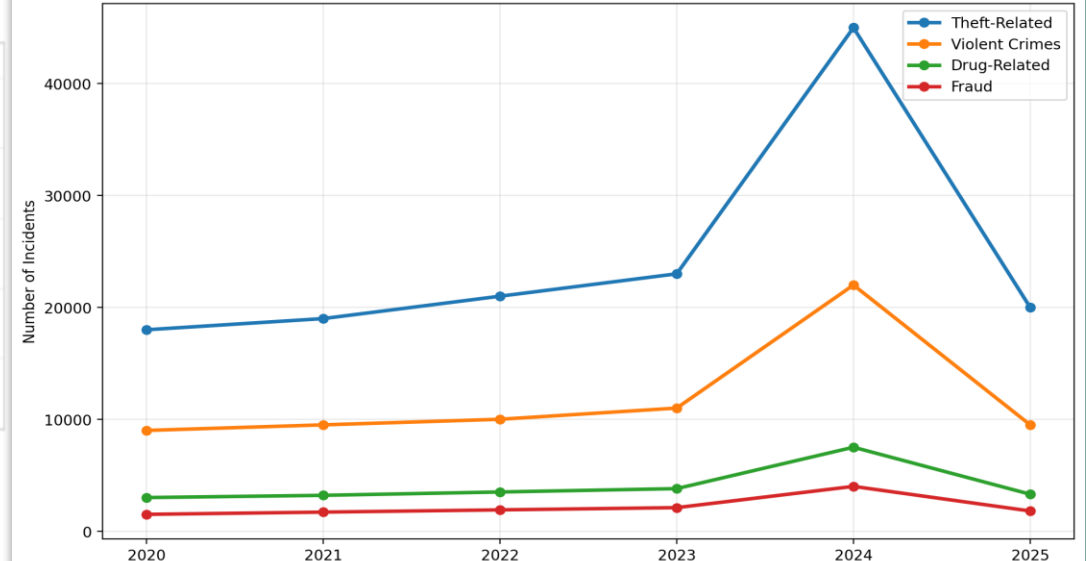
Primary Drivers of the 2024 Anomaly:

- A significant surge in **violent crime, particularly gun violence**.
- **Organized crime activity** driving nationwide increases in auto theft.
- **Ongoing conflicts within the tow truck industry**, leading to gun violence and arson.
- Proliferation of **illegal firearms**.
- Challenges related to the national bail system for firearm offences.

Implication:

These drivers converged to create a 'storm,' overwhelming existing systems and exposing critical vulnerabilities in our resource allocation, particularly in District 4.

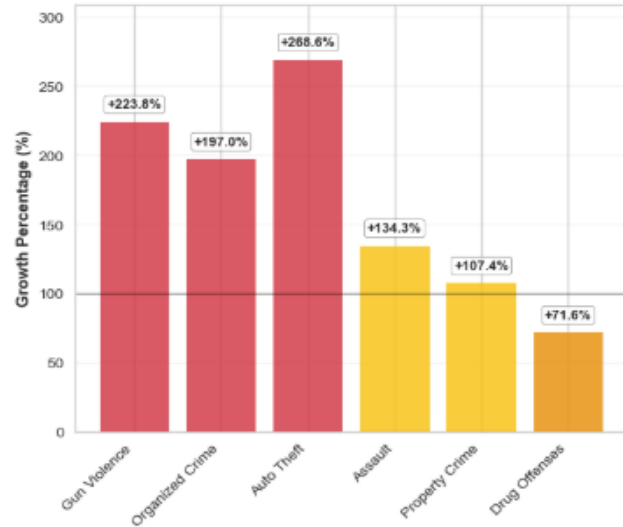
Crime Type Trends Over Time



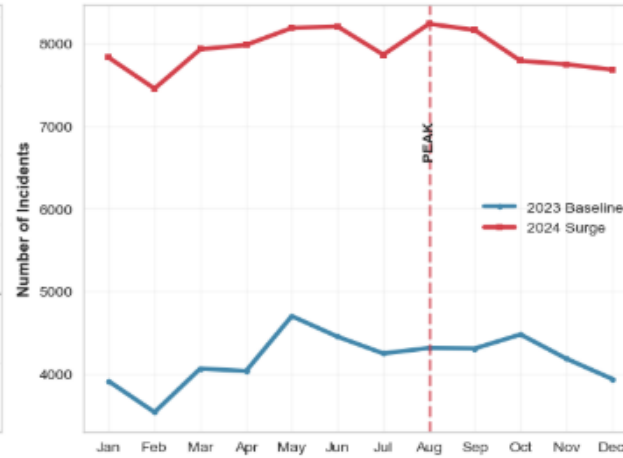


2024 Crime Surge: Key Drivers

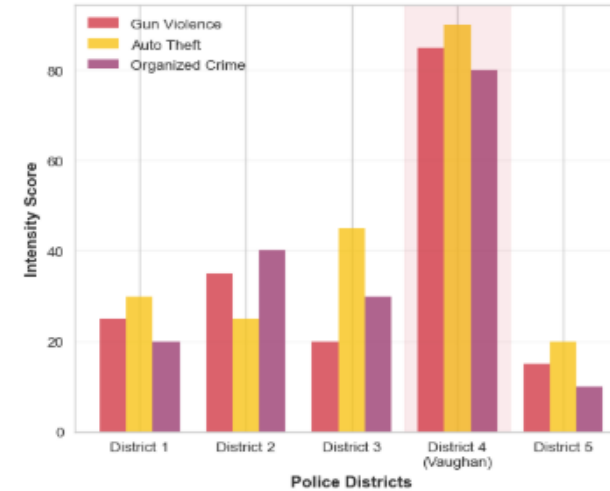
YEAR-OVER-YEAR GROWTH BY CATEGORY
2024 vs 2023



MONTHLY INCIDENT VOLUME COMPARISON
2023 vs 2024



CRIME DRIVER INTENSITY BY DISTRICT
District 4 (Vaughan) Identified as Hotspot



Key Drivers:

- Auto Theft: +268.6%
- Gun Violence: +223.8%
- Organized Crime: +197.0%
- Assault: +134.3%
- Property Crime: +107.4%
- Drug Offenses: +71.6%

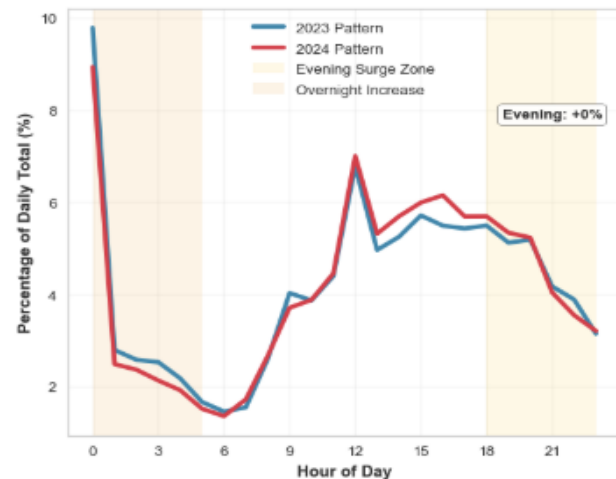
Geographic Hotspot:

- District 4 (Vaughan): EPICENTER

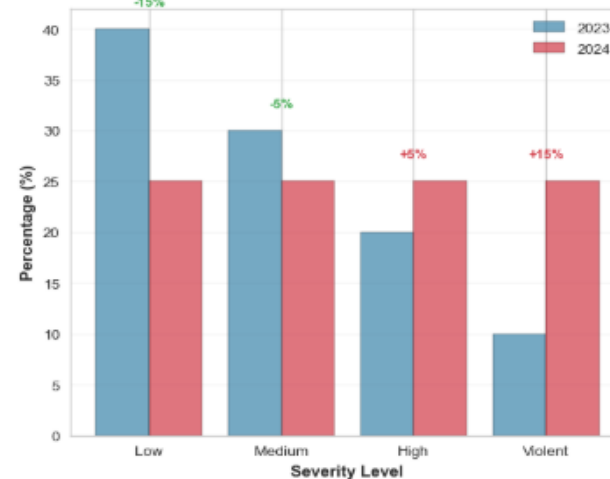
Strategic Implications:

- **Immediate Deploy** 60% of new tactical unit capacity and 50% of organized crime investigative resources to District 4 hotspots
- **Timing:** Enhance evening overnight patrol coverage by 40%
- **Targeting:** Disrupt organized auto theft networks as priority
- **Planning:** Implement summer surge response plan for 2025
- **Prevention:** Address root causes of gun violence spike

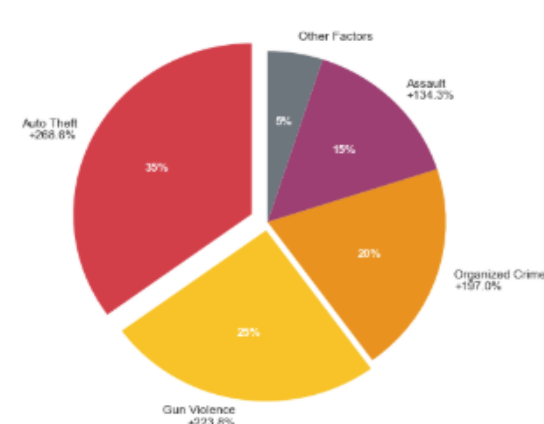
HOURLY DISTRIBUTION SHIFTS
Temporal Pattern Changes



CRIME SEVERITY DISTRIBUTION SHIFT
Increased High-Severity Incidents

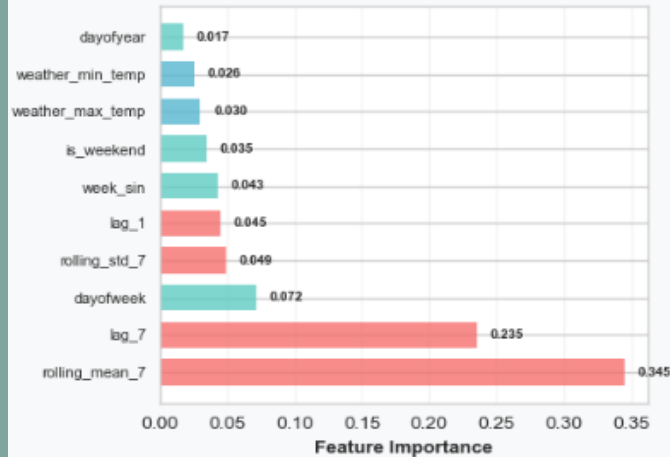


PRIMARY DRIVERS OF 2024 SURGE
Weighted by Growth Magnitude

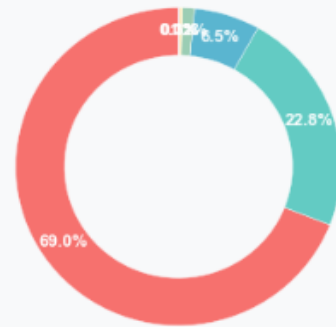


Crime Predict Driver Factors

TOP 10 PREDICTIVE FACTORS



IMPORTANCE BY CATEGORY



Top Predictive Factors

- Count from the Same Month in Previous Year 68.3%
- Date of the week 18.2%
- Average temperature 6.5%

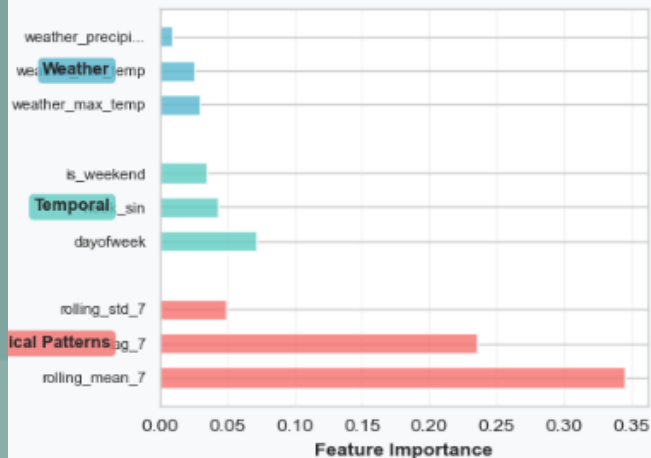
Recommended Next Steps

- Focus on the top 6 factors
- Implement rolling 7-day monitoring
- Develop weekly pattern analysis for different crime types

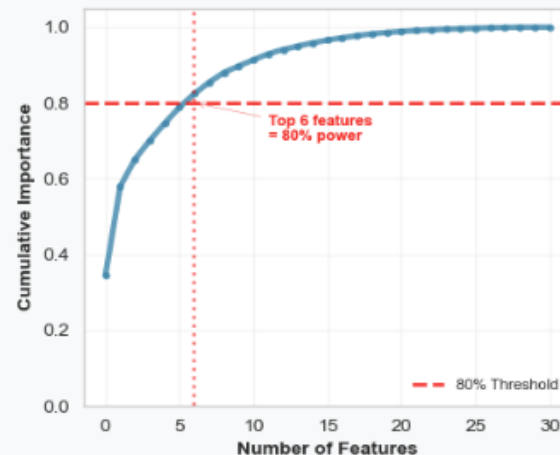
Enhancement Opportunities

- Add holiday and special event indicators
- Incorporate economic indicators (unemployment, etc.)
- Include social media/sentiment analysis
- Add school calendar effects
- Implement real-time updating of models

TOP FACTORS BY CATEGORY



CUMULATIVE PREDICTIVE POWER



Concluding Insight

- This analysis confirms that while historical patterns are predictable, the 2024 surge was driven by factors outside these models (like organized crime), justifying our need for both predictive resource planning and targeted tactical interventions



Multi-Sources Crime Intelligence

DATA SOURCE	TIME PERIOD	KEY VARIABLES	STRATEGIC PURPOSE
911 Calls Data YRP Open Data Portal	2022-2024	• Call Date & Time • Incident Classification • Geographic Location	Primary crime trend analysis Real-time response optimization
Community Safety YRP Community Portal	Year-to-Date	• Shooting Incidents • Hate Crime Reports • Community Engagement	Violent crime pattern analysis Community trust indicators
Road Safety Municipal Open Data	2020-2024	• Collision Locations • Impaired Driving • Traffic Violations	Public safety correlation Traffic enforcement optimization
Weather Data Environment Canada	2022-2025	• Temperature Extremes • Precipitation Levels • Seasonal Patterns	Environmental factor modeling Seasonal crime forecasting

Data Quality & Integration

Analytical Advantage

Predictive Power Analysis

DATA SOURCE	QUALITY	STATUS	DATA CATEGORY	IMPORTANCE	KEY INSIGHTS	METRIC	TRADITIONAL	INTEGRATED	GAIN
911 Calls Data	95%	Core integrated	Historical Patterns	68.3%	• Weekend crime peaks • Summer surge patterns • Monthly cycles	Predictive Accuracy	Single-source	Multi-source	↑ 25%
Weather Data	100%	API complete	Community Safety	12.1%	• Shooting clusters • Hate crime correlation • Community sentiment	Resource Efficiency	Reactive	Proactive	↑ 40%
Road Safety	85%	Spatial mapping	Road Safety	8.2%	• Accident-crime hotspots • DUI-violence correlation • Traffic risks	Risk Identification	Basic mapping	100-pt scoring	Early warning
Community Safety	90%	Categorical mapping	Weather Factors	6.5%	• Temperature correlation • Precipitation impact • Seasonal effects	Strategic Planning	Short-term	Long-term focus	Dual strategy

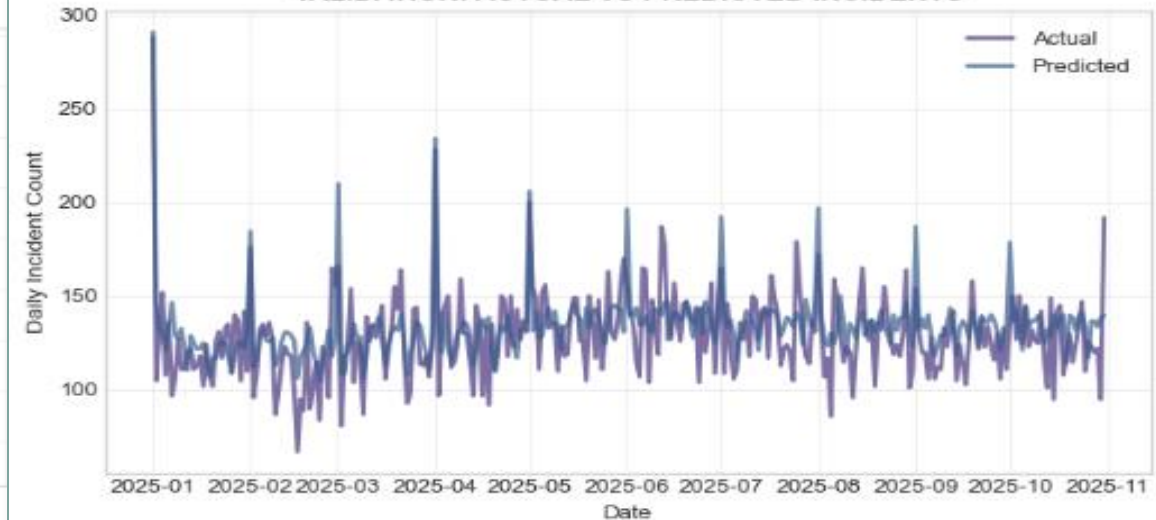
Crime Time-Series Modeling

ENHANCED CRIME PREDICTIONS WITH UNCERTAINTY QUANTIFICATION
November 2025 - December 2026



- **Data:** Training: 2020 to 2023; Test: Jan 2025 to Oct 2025; Outlier: 2024
- **Model & Performance:** Random Forest: MAE=15.1, RMSE=18.9, $R^2=0.32$
Gradient Boosting: MAE=15.4, RMSE=19.4, $R^2=0.29$
Ensemble (weighted (RF: 0.53, GB: 0.47)): MAE=15.0, RMSE=18.9, $R^2=0.32$
- **The 'Low' R^2 :** Our model captures predictable seasonal patterns for staffing. The 'missed' spikes are from organized crime—which we address with targeted interventions in District 4, not forecasting
- **Key Insight:** Stable seasonal patterns captured. While our model reliably predicts the seasonal ebb and flow of crime (e.g., more in summer, less in winter), it confirms that major crime surges are driven by complex, unpredictable factors like organized crime spikes. This is why our strategic plan combines this baseline forecast with a targeted intervention in District 4
- **Strategic Recommendation:** Resources Planning: Align with seasonal patterns

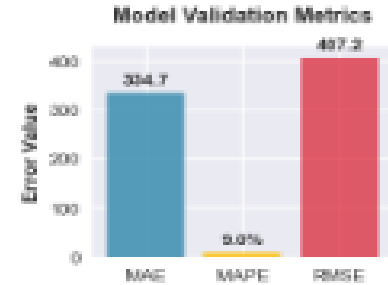
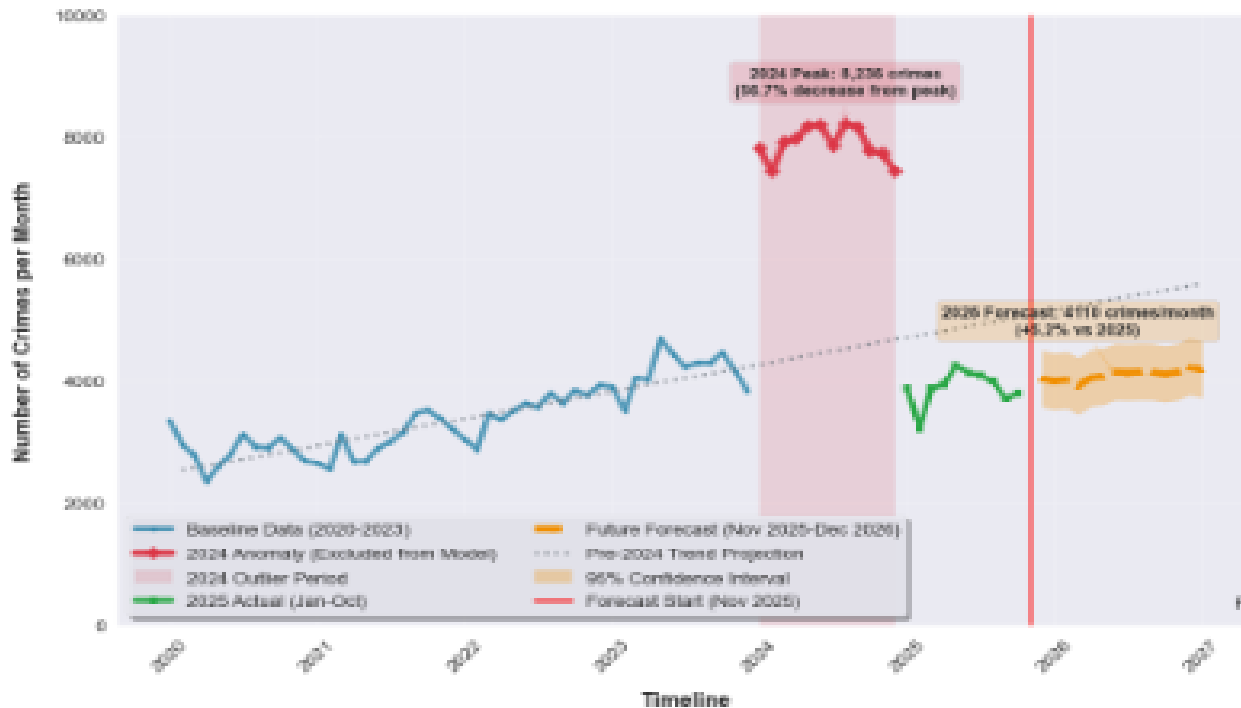
VALIDATION: ACTUAL VS PREDICTED INCIDENTS



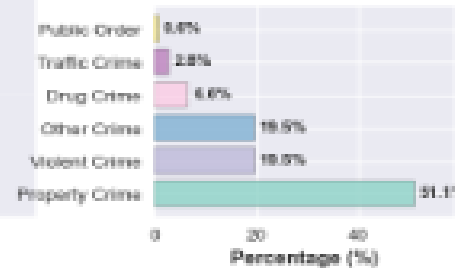


Forecast: A Return to Growth in 2026 Demands Action

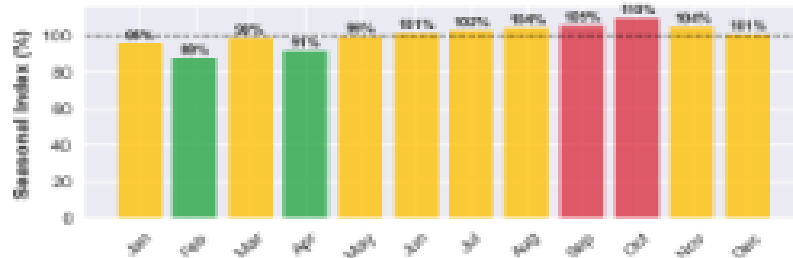
York Region Crime Forecast: 2020-2026



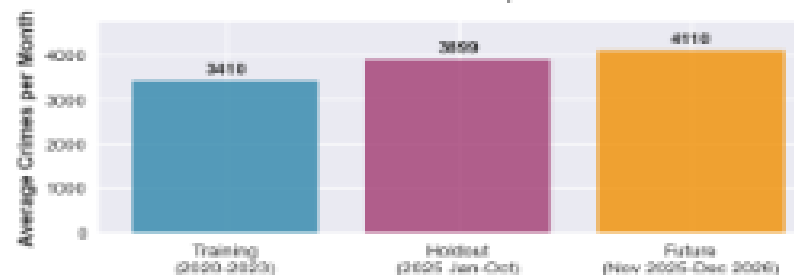
Crime Category Distribution (Training Period 2020-2023)



Seasonal Crime Patterns (Percentage of Monthly Average)



Crime Rate Comparison



Forecast

While crime levels are projected to remain well below the 2024 peak, our forecasts indicate a return to a growing trend in 2026, necessitating proactive measures

Strategic Recommendations

1. Base 2026 planning on 3394 crimes/month forecast
2. Maintain ± 502 crimes flexibility in budgets
3. Monitor category-specific trends for targeted interventions

Priority

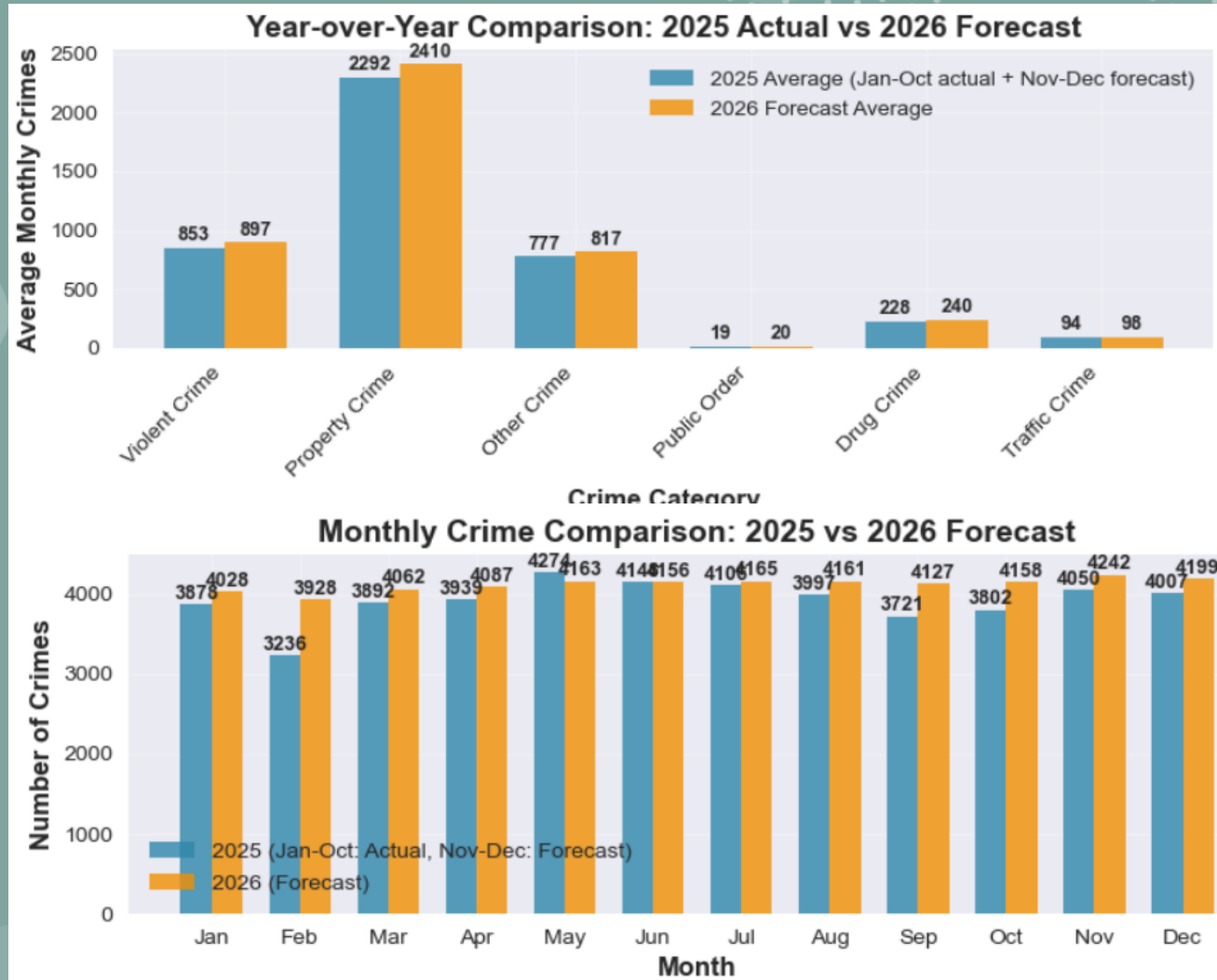
Implement strategic shift bidding: Prioritize full staffing on Mondays-Thursdays to cover our vulnerability period, using overtime strategically for Friday peak activity. **Why focus on weekdays?** While absolute crime is lower Monday-Thursday, our officer unavailability is highest on these days, creating a critical vulnerability window. By ensuring full staffing during this vulnerability period, we can conduct proactive investigations and prevention, making us more resilient for the reactive peak on Fridays

Action

Activate the 'Summer Surge Plan' from June 1st, pre-deploying tactical units to District 4 hotspots

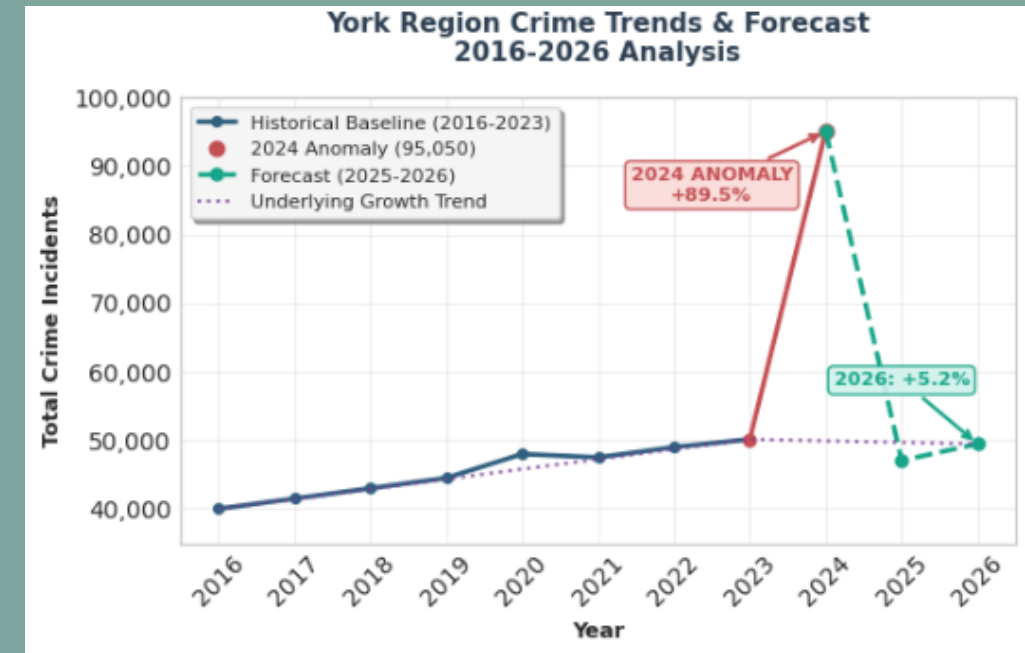


Forecasted Increase And Urgent Action Needed



Urgent Action Needed

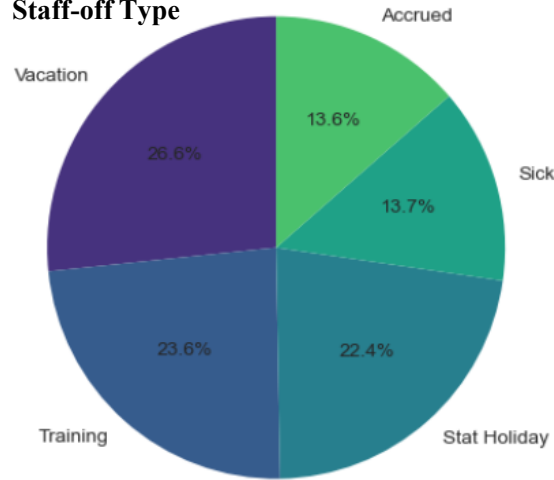
- Priority: Proactively reallocate resources to counter the forecasted 5.2% increase in crime for 2026
- The 2024 surge was a unique, extreme event. Our forecast shows a return to the underlying pre-2024 growth trend, indicating that the systemic pressures that existed before the 2024 spike are still present and will lead to increased crime again if unaddressed



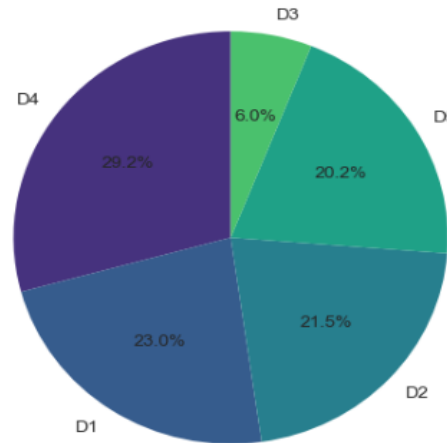


District 4 (Vaughan) Operations & The Staffing Crisis

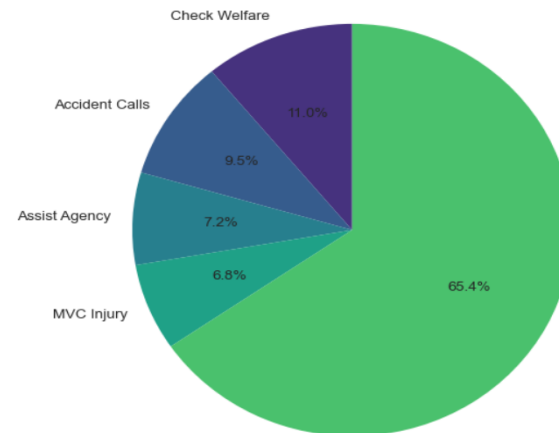
Staff-off Type



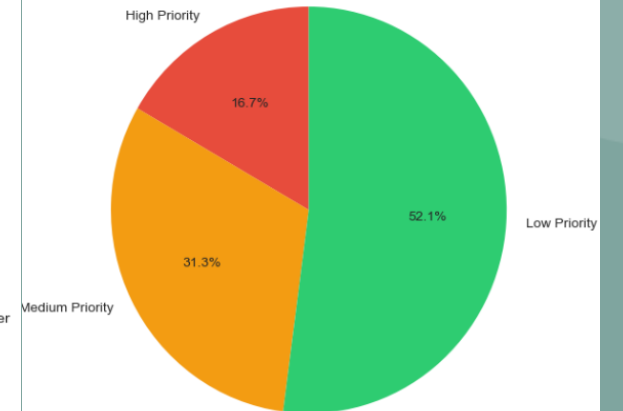
District 4: 29% of All Calls



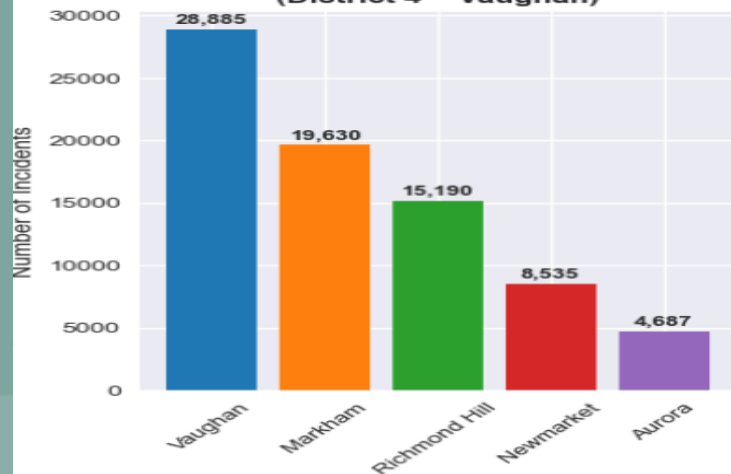
District 4 Call Types



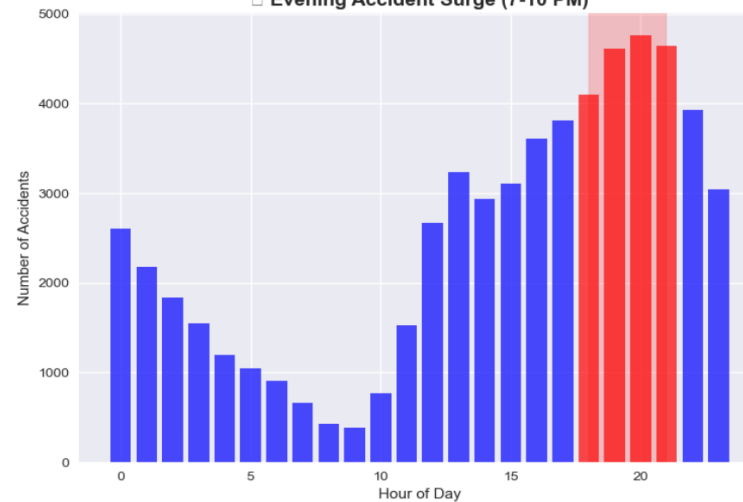
Call Priority Distribution



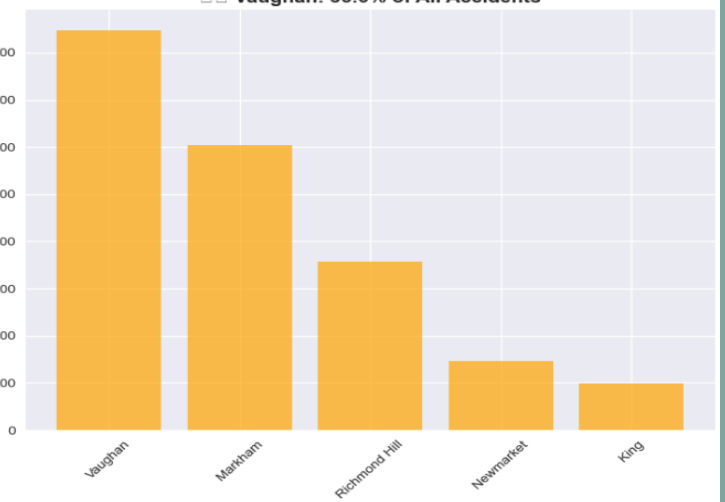
Crime by Municipality (District 4 = Vaughan)



Evening Accident Surge (7-10 PM)



Vaughan: 35.6% of All Accidents





Top 10 High-Impact Operational Days

District 4 Staffing Risk Analysis

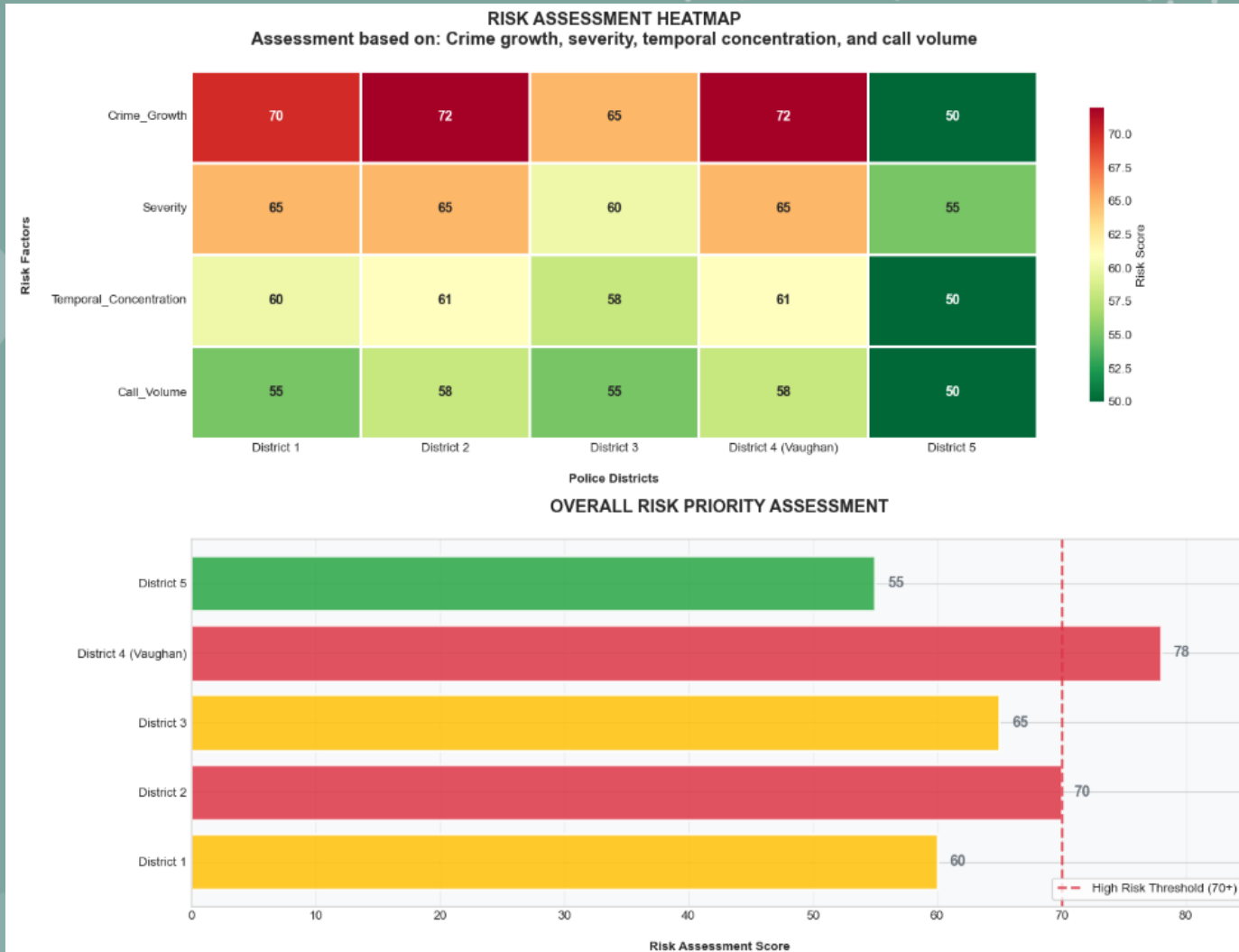
Rank	Date	Day	Officers Off	Calls	Officers On-Duty	Officer Deficit
1	2024-09-13	Friday	45	546	16	YES
2	2024-09-12	Thursday	47	515	14	YES
3	2023-10-12	Thursday	46	526	15	YES
4	2024-09-10	Tuesday	45	518	16	YES
5	2024-12-13	Friday	41	556	20	YES
6	2024-12-04	Wednesday	37	607	24	NO
7	2024-09-11	Wednesday	42	531	19	YES
8	2024-12-05	Thursday	41	542	20	YES
9	2024-06-20	Thursday	42	526	19	YES
10	2024-11-12	Tuesday	40	548	21	YES

Key Insights

- 9 out of 10 highest-risk days have officer deficits
- September 2024 shows cluster of high-risk days
- Minimum staffing requirement: 23 officers



District 4 (Vaughan) is the Epicenter of This Risk



Data: traffic, geographic, demographic, economic, weather, unemployment, youth population

Model: XGBoost for feature importance

Priority Assessment

- 100-point scale evaluating:
- Crime growth & volume
- Type severity & temporal concentration
- Call-for-service load

Tier Definition:

- High (70+): Critical risk requiring immediate intervention
- Medium (60-69): Elevated risk requiring monitoring and prevention
- Low (<60): Stable, standard patrol levels sufficient

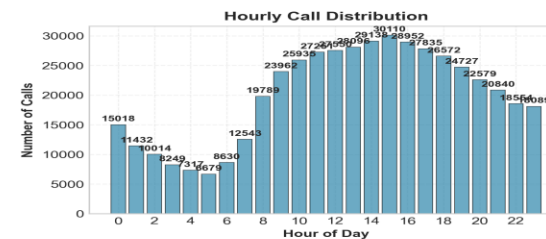
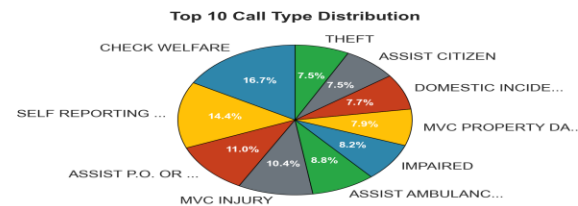
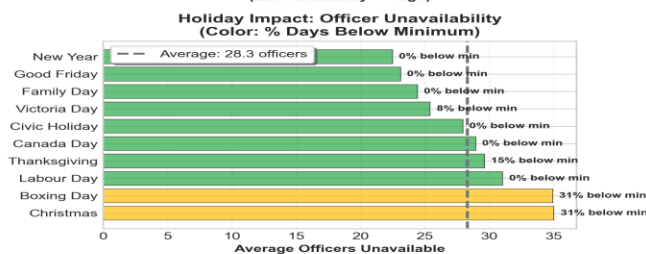
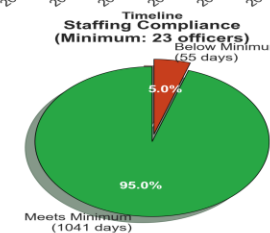
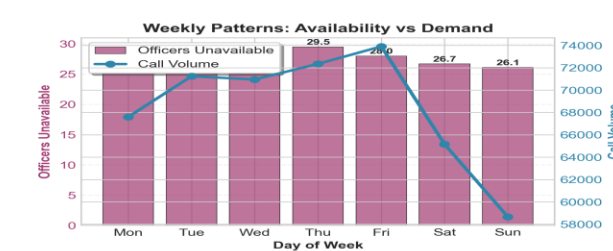
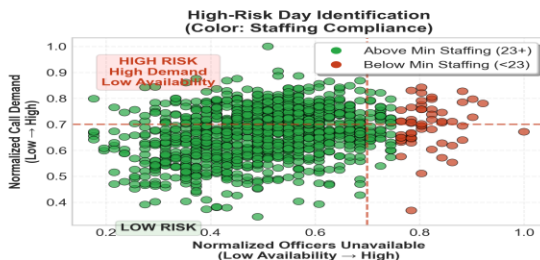
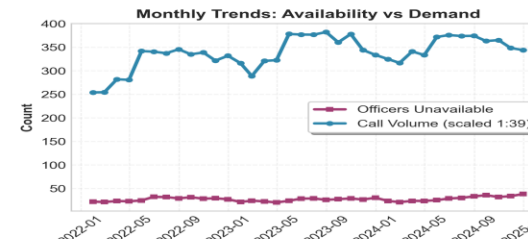
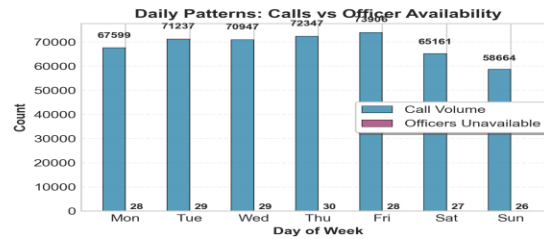
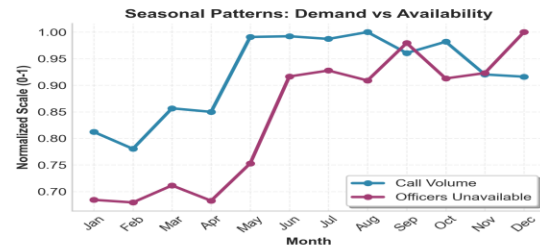
Key Insights

- This scoring system allows us to move beyond simple crime counts and focus resources on areas with the most severe and concentrated threats, such as District 4
- The crisis in District 4 is not just high crime volume, but a dangerous mismatch between criminal activity patterns (evenings, weekends) and officer availability (weekday vulnerability)
- The data confirms that District 4 is not just a hotspot, but a systemic outlier requiring a disproportional share of resources



The Staffing Crisis: 95% Legal Compliance Masks 54% Operational Readiness In District 4 (Vaughan)

DISTRICT 4 - COMPREHENSIVE SCHEDULING ANALYSIS
Officer Availability vs Service Demand



Key Performance Indicators Summary

Overall Performance: GOOD	
Staffing Compliance: 95.0%	
Avg Officers Unavailable	51.0
Max Officers Unavailable	33.0
Avg Officers Available	33.0
Days Below Min (23)	55.0
Staffing Compliance Rate	95.0%
Total Officers (Est)	61.0
Total Calls	479,861
Avg Daily Calls	438
Call Processing Days	1096

Summary

- Officer Readiness: 33 / 61 (54%)
- Minimum Required Officers: 23
- Staffing Compliance Rate: 95% → **Misleading**
- ❖ While we meet a bare-minimum legal requirement 95% of the time, our effective operational capacity is only 54% due to vacancies, leave, and training. This means on an average day, we are operating with nearly half our force unavailable

High Risk Periods

- High-Unavailability Months: August, September and October
- High-Unavailability Days: While criminal activity peaks on Fridays, our operational vulnerability is highest Monday-Thursday due to low officer availability. This mismatch requires strategic scheduling adjustments



Conclusion & Strategic Recommendations

- **District 4 (Vaughan) is the Epicenter of Risk**
- **Immediate Tactical Action:** Deploy resources to District 4 to combat organized crime and gun violence
- **Proactive Staffing Strategy:** Implement strategic shift bidding to cover weekday vulnerability and the Summer Surge Plan
- **Long-Term System Optimization:** Use the forecasted 2026 trend to advocate for sustained investment in prevention and data monitoring



Thank You and Q&A

- **Download this Presentation**

Download Full Presentation:



Or visit: https://github.com/fuxima/YRP/blob/main/YorkRegionCrimeAnalysis_presentation.pptx

- **View Full Project and Code**

View Project on GitHub:



Or visit: <https://github.com/fuxima/YRP>

- **Contact:** Email: fuxi.ma@outlook.com



About Myself

Canadian Citizen & PhD in Applied mathematics

- Since about 20 years ago when immigrated to Canada

Career (Total Experience 15 Years)

- Engineering (Climate change and sea-level rise predictive modeling)
- Pharmaceutical (Clinical trial data & patients data modeling)

Data Science – ML/AI

- Python, R, SQL, PySpark
- Models: Traditional ML: Regression, Classification, Clustering, Xgboost, LightGBM
Deep Learning: RNN, TensorFlow/Pytorch & Keras, LSTM for time series data
- Platform: Databricks, Cloudera, AWS & Azure
- Data Visualizations: Power BI, Spotfire
- Dashboard: R-Shiny, Flask, Dash, Streamlit
- Report/Documentation: Quarto



Climate Change and Sea Level Rise Prediction

Time-series forecast

- ETL Data
- EDA
- Output
 - Joint Probability (Storm Surge + Seal Level Rise in Bay of Fundy)
 - <https://fuxima.wixsite.com/fishway/about-form>



APP: Rx Prescription Data, Patient Data, Claim Data & Drug Store Universe Data

Data Ingestion: Collect data from All States in USA & Prediction Modeling
Anomaly Detection

- Moving Average $\pm 3\sigma$; Box plot – IQR; Z-score

Holiday Impact

- Holiday drop; Holiday re-bounce

Model: TensorFlow + Keras; LSTM

Hyperparameter Tuning

- Different layers; Learning Rate; Dropout Rates;
Early Stopping; Embedding Dimensions

Generate Weekly Report in word/pdf/html

Deliverables (offering) for Pharma, Payers, Providers, Government

[Key Driver Analysis](#)



Deep Learning (RNN) – Cancer Prediction

Model

- TensorFlow + Keras
- LSTM

Patient Sequential Data

- Tokenization
- Embedding

Imbalanced Classes

- Class Weighting
- SMOTE (Synthetic Minority Over-sampling Technique)

Hyperparameter Tuning

- Different layers
- Dropout rates
- Embedding dimensions
- Learning rate

[Dashboard](#)



Canadian Education and Labour Market

Dashboard

 **NGS2020 Database Analyzer**

Table Selection

🔍 Search tables:

Type to search tables by name or description...

📊 Select table (113 available):

AFT_050 - After 2020 program - Full-time or part-time - 1st job or business

AFT_070 - After 2020 program - Permanent/not permanent - 1st job or business

AFT_080 - After 2020 program - Reason job not permanent - 1st job or business

AFT_090 - After 2020 program - Relatedness of 1st job/business to program

AFT_P010 - After 2020 program - Number of jobs or businesses

AFT_P020 - After 2020 Program - Length of time until first job or business

AFT_P040 - After 2020 program - Employee or self-employed - 1st job or business

BEF_160 - Before 2020 program - Number of months of work experience

 **Analyze Table**



Data Integration & Modeling

Data Polling

- Ten Neuroscience Studies

Challenge

- Variety of Data Structures and Format
- Data harmonization (Using SAS, SQL converted the source data into R)

R Shiny App

- Integrated Summary of Safety (ISS)
- Integrated Summary of Efficacy (ISE)
- Cross-study meta-data modeling
 - Patients Enrolment
 - Patient Dropout

This Approach Applies to YRP's Multi-Sources Crime Intelligence



AI Agents: Generative AI & LLM Integration

- Developed production applications using ChatGPT/DeepSeek, LangChain, and Python (RESTful APIs: official SDKs and direct HTTP calls)
- Implemented text-generation pipelines, conversational memory, and RAG (Retrieval-Augmented Generation). I used vector databases (ChromaDB, Pinecone).
- Created agents for LLMs to access our different databases, do analysis and deliver summary including FDA Announcement, competitor press releases, market reports, Natural events (Stm/HURC) and Holiday impacts including Holiday Drop and post-holiday rebound.



Recall is the percentage of a class correctly predicted for that class

Recall (also known as sensitivity or **true positive rate**) measures the proportion of actual positive cases that were correctly identified by the model

$$\text{Recall} = \text{True Positives} / (\text{True Positives} + \text{False Negatives})$$

$$\text{Precision} = \frac{\text{correctly classified actual positives}}{\text{everything classified as positive}} = \frac{TP}{TP + FP}$$

$$\text{F1 score} = 2 * (\text{Precision} * \text{Recall}) / (\text{Precision} + \text{Recall})$$

F1 Score is the harmonic mean of recall and precision

		Predicted Class		
		Positive	Negative	
Actual Class	Positive	True Positive (TP)	False Negative (FN) Type II Error	Sensitivity $\frac{TP}{(TP + FN)}$
	Negative	False Positive (FP) Type I Error	True Negative (TN)	Specificity $\frac{TN}{(TN + FP)}$
		Precision $\frac{TP}{(TP + FP)}$	Negative Predictive Value $\frac{TN}{(TN + FN)}$	Accuracy $\frac{TP + TN}{(TP + TN + FP + FN)}$



1. "Your R^2 is quite low (0.32). How can we trust a model with such low explanatory power?"

Answer: "That's an excellent question. The R^2 measures how well the model explains the *variance* in the data. Our model is very good at capturing the predictable, seasonal baseline—the 'heartbeat' of crime for resource planning. The 'missed' variance are the large, unpredictable spikes from events like organized crime rings, which are not easily forecast with traditional time-series data. This is precisely *why* our strategy is two-pronged: use the model for baseline staffing and use tactical intelligence for the spikes. A perfect model for crime is unrealistic; a smart strategy that understands model limitations is essential."

2. "You integrated external data. Can you walk us through your feature selection process and why you chose those specific datasets?"

- **Answer:** "Certainly. I focused on factors with a plausible, documented correlation to crime or police workload. I used **weather data** (temperature), as higher temperatures can correlate with increased outdoor activity and certain crime types. More critically, I incorporated **holiday calendars and road safety data** to account for predictable surges in non-crime call volumes (like traffic accidents) that still tie up officer resources. For feature selection, I used models like XGBoost to calculate feature importance, ensuring the external data added predictive value."

3. "How would you improve this model if you had more time and resources?"

"I would focus on three areas:

1. **Data Enrichment:** Integrate more granular data, like real-time economic indicators (unemployment filings) or anonymized mobility data.
2. **Model Sophistication:** Experiment with more advanced deep learning models like LSTMs or Transformers that might better capture complex, long-range dependencies.
3. **Operational Integration:** Build a real-time dashboard that updates the forecast weekly or daily, allowing for dynamic resource allocation, and use LLM agents to automatically scan news and social media for early signals of emerging tensions."

4. "Your plan calls for shifting resources to District 4. What do you say to the Commanders of other Districts who will see this as robbing Peter to pay Paul?"

- **Answer:** "That's a critical operational concern. The data shows District 4 is a systemic outlier. This isn't about favoring one district but about deploying resources to where the threat is most concentrated and severe to protect the *entire region*. A major crime event in District 4, like an organized auto-theft ring or a spike in gun violence, often drains resources from surrounding districts anyway. By concentrating efforts to 'put out the biggest fire,' we stabilize the entire region. This should be presented as a temporary, data-supported surge to address a critical vulnerability."

5. "How would you implement the 'Strategic Shift Bidding' in practice, considering union contracts and officer morale?"

- **Answer:** "I recognize this is as much a human challenge as a technical one. I would propose a pilot program framed as a 'Proactive Patrol Initiative.' The data shows that ensuring full staffing on weekdays reduces burnout by preventing reactive chaos on Fridays. We could use incentive-based overtime for Mondays-Thursdays, rather than forced reassignments. The key is to present the data to the officers and their representatives, showing them how this plan actually creates more predictable schedules and reduces their stress during high-chaos periods."



Behavioral & Role-Fit Questions

6. "This role involves explaining complex data to non-technical audiences, including senior officers. How do you do that?"

Answer: "I focus on the 'so what?' and avoid technical jargon. For example, instead of saying 'the XGBoost model identified a high feature importance for the temporal variable,' I would say, 'the data tells us that historical patterns from last year are the single biggest predictor of crime this month.' I use clear, intuitive visuals like the ones in this presentation—hotspot maps, simple bar charts for growth rates, and the '95% vs 54%' comparison—to make the insight immediate and undeniable."

7. "Where do you see the future of AI and Data Science in policing, and how do you fit into that?"

- **Answer:** "The future is in **predictive prevention and intelligent automation**. Beyond forecasting, AI can help in cybercrime detection, analyzing complex networks of organized crime from text reports, and optimizing patrol routes in real-time. My experience, especially with building LLM applications, positions me to help YRP not just analyze the past, but to build the tools that will give our officers a decisive advantage in the future. I'm not just a data scientist; I'm a builder of solutions that enhance public safety."

On Collaboration: "The job description mentions close collaboration with GIS, Data Engineering, and front-line decision-makers. Could you describe what a typical collaborative project looks like from initiation to deployment? What does a successful partnership look like in practice?"

On the Tech Stack Evolution: "I noticed the presentation mentioned tools like Python, SQL, and Power BI. Could you describe the current on-premise or cloud data infrastructure (e.g., Azure, AWS)? Are there any planned investments or shifts in the technology roadmap for the bureau?"

On MLOps & Model Deployment: "The purpose of this role is to transform data into strategic assets that support decision-making. Could you talk about the process for taking a model from a Jupyter Notebook to a production environment? Is there a established MLOps pipeline, or is that an area for development?"

On the AI Strategy: "York Regional Police seems committed to becoming more data-driven. Is there a formal AI or data strategy in place? How does leadership envision the role of advanced analytics, like the LLM applications I mentioned, evolving in the next few years?"